13.6.5 Wrap-Up: Growth Mindset

1. Complete the table.

Which mindset did I demonstrate?	Check the best description.
Did I use the partner discussions as learning opportunities?	○ Never ○ Sometimes ○ Usually ○ Always
Did I work on problems that challenged me?	○ Never ○ Sometimes ○ Usually ○ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	○ Never ○ Sometimes ○ Usually ○ Always

Unit 13, Lesson 7: Exploring Transformations by a Factor

Benchmark

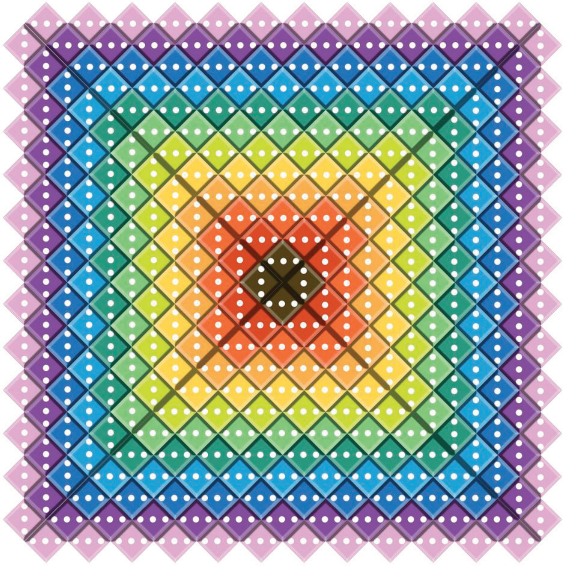
MA.912.F.2.1 Identify the effect on the graph or table of a given function after replacing $f(x)$ by $f(x)+k$, $kf(x)$, $f(kx)$, and $f(x+k)$ for specific values of k.

Learning Targets

- ☐ I can discover the effect on the graph of $f(x)$ for a transformation of $f(kx)$ or $kf(x)$.
- ☐ I can discover the effect on the table of values of $f(x)$ for a transformation of $f(kx)$ or $kf(x)$.

13.7.1 Warm-Up: Wonder and Notice

A picture of dice is shown.



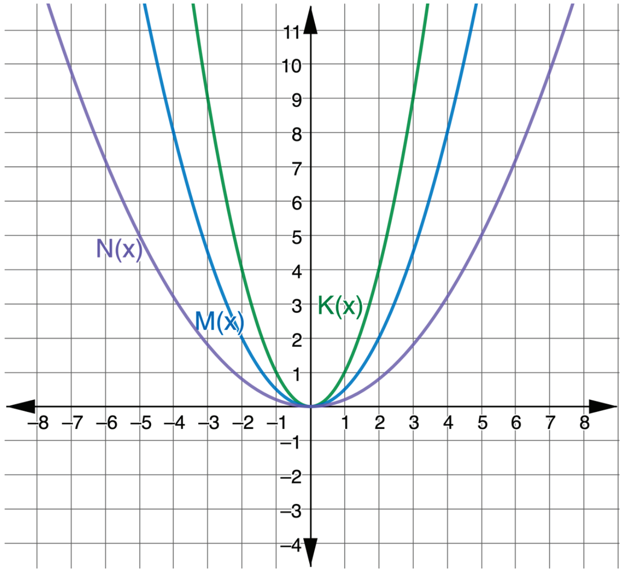
1. What do you notice?

2. What do you wonder?

3. Write a question about the picture that you could ask a 5th grade student.

13.7.2 Exploration: Transformations by a Factor

1. The functions $K(x)$, $M(x)$, and $N(x)$ are shown on the graph where $\boldsymbol{K(x) = x^2}$, $\boldsymbol{M(x) = \frac{1}{2}x^2}$, and $\boldsymbol{N(x) = \frac{1}{5}x^2}$. The table of values for each function are also shown.



$y = K(x)$	
x	y
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9

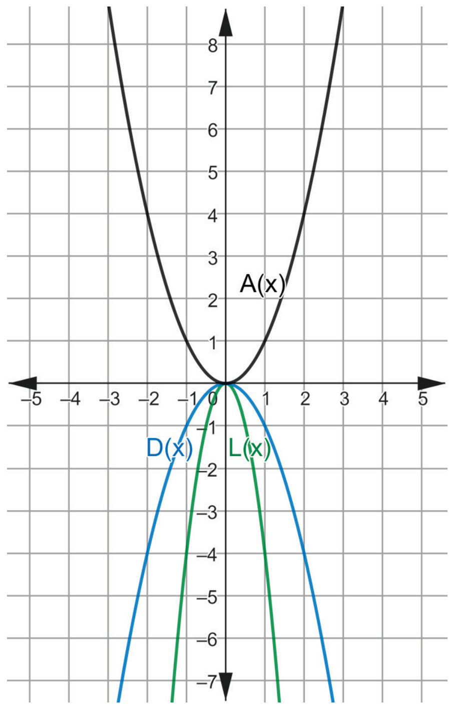
$y = M(x)$	
x	y
-3	$\frac{9}{2}$
-2	2
-1	$\frac{1}{2}$
0	0
1	$\frac{1}{2}$
2	2
3	$\frac{9}{2}$

$y = N(x)$	
x	y
-3	$\frac{9}{5}$
-2	$\frac{4}{5}$
-1	$\frac{1}{5}$
0	0
1	$\frac{1}{5}$
2	$\frac{4}{5}$
3	$\frac{9}{5}$

Discuss with your partner the effect the coefficient has on the y-values. Then, complete the summary.

The coefficient of $M(x) = \frac{1}{2}x^2$ is _______. The y-values of $M(x)$ can be found by multiplying the y-values of $K(x)$ by ________ for each corresponding x-value. The coefficient of $N(x) = \frac{1}{5}x^2$ is _______. The y-values of $N(x)$ can be found by multiplying the y-values of $K(x)$ by ________ for each corresponding x-value.

2. The functions $\boldsymbol{A(x)}$, $\boldsymbol{D(x)}$, and $\boldsymbol{L(x)}$ are shown on the graph, where $\boldsymbol{A(x) = x^2}$, $\boldsymbol{D(x) = -x^2}$, and $\boldsymbol{L(x) = -4x^2}$. The table of values for each function are also shown.

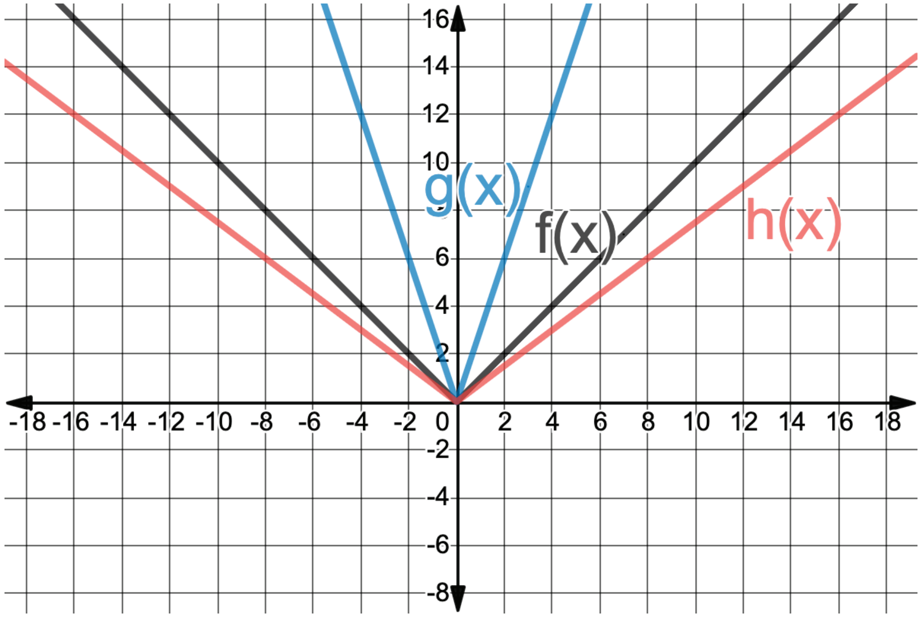


$y = A(x)$		$y = D(x)$		$y = L(x)$	
x	y	x	y	x	y
-3	9	-3	-9	-3	-36
-2	4	-2	-4	-2	-16
-1	1	-1	-1	-1	-4
0	0	0	0	0	0
1	1	1	-1	1	-4
2	4	2	-4	2	-16
3	9	3	-9	3	-36

Discuss with your partner the effect the coefficient has on the y-values. Then, complete the summary.

The coefficient of $D(x) = -x^2$ is _______. The y-values of $D(x)$ can be found by multiplying the y-values of $A(x)$ by ________ for each corresponding x-value. The coefficient of $L(x) = -4x^2$ is _______. The y-values of $L(x)$ can be found by multiplying the y-values of $A(x)$ by ________ for each corresponding x-value.

3. The functions $f(x)$, $g(x)$, and $h(x)$ are shown on the graph where $f(x) = |x|$, $g(x) = |3x|$, and $h(x) = |0.75x|$. The table of values for each function are also shown.



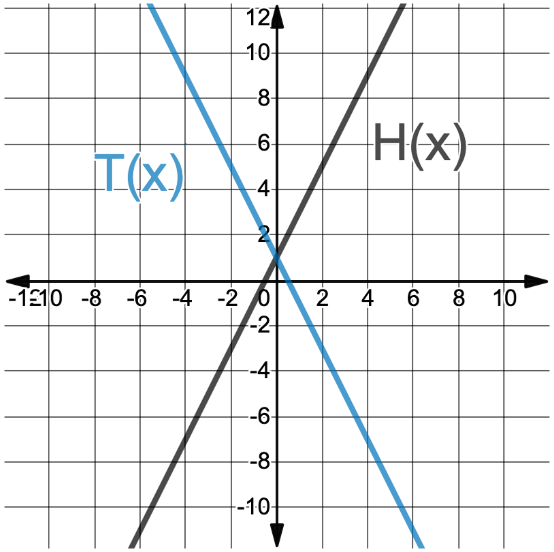
$y = f(x)$		$y = g(x)$		$y = h(x)$	
x	y	x	y	x	y
-3	3	-1	3	-4	3
-2	2	$-\frac{2}{3}$	2	$-2\frac{2}{3}$	2
-1	1	$-\frac{1}{3}$	1	$-\frac{4}{3}$	1
0	0	0	0	0	0
1	1	$\frac{1}{3}$	1	$\frac{4}{3}$	1
2	2	$\frac{2}{3}$	2	$2\frac{2}{3}$	2
3	3	1	3	4	3

Discuss with your partner the effect the coefficient has on the table of values. Then, complete the table.

Function	Values that Changed from $f(x) = \|x\|$	Description of the Change
$g(x) = \|3x\|$	○ x-values ○ y-values	○ Divided by _____ ○ Multiplied by _____
$h(x) = \|0.75x\|$	○ x-values ○ y-values	○ Divided by _____ ○ Multiplied by _____

4. The functions $H(x)$ and $T(x)$ are shown on the graph where $T(x) = H(-x)$. The table of values for each function are also shown.

$y = H(x)$		$y = T(x)$	
x	y	x	y
-3	-5	-3	7
-2	-3	-2	5
-1	-1	-1	3
0	1	0	1
1	3	1	-1
2	5	2	-3
3	7	3	-5



a. Discuss with your partner the effect on the table of values when the x in $H(x)$ was replaced with $-x$ to create $T(x)$.

b. Discuss with your partner the effect on the graph when the x in $H(x)$ was replaced with $-x$ to create $T(x)$.

13.7.3 Bring It Together: The Effect of Transformations of Factors

1. Complete the table by determining which values in the table of values of $f(x)$ change, then describe the effect of the factor.

Function Notation	k-values	Which Values Change in the Table of Values?	Effect of the Factor
$f(kx)$	$k < -1$	○ x-values ○ y-values	
$f(kx)$	$-1 < k < 1,$ $k \neq 0$	○ x-values ○ y-values	
$f(kx)$	$k > 1$	○ x-values ○ y-values	
$kf(x)$	$k < -1$	○ x-values ○ y-values	
$kf(x)$	$-1 < k < 1$	○ x-values ○ y-values	
$kf(x)$	$k > 1$	○ x-values ○ y-values	

13.7.4 Guided Practice: Describing Transformations

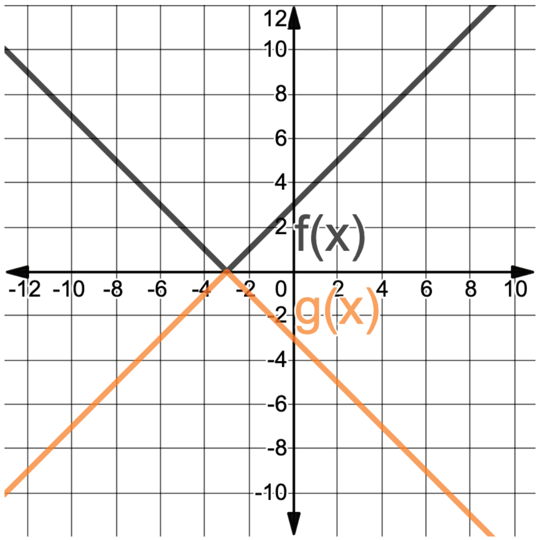
1. The table of values of the function, $y = W(x)$, is shown.

x	-9	-6	-3	0	3	6	9
W	-12	-10	-8	-6	-4	-2	0

Complete the sentence to describe the effect on the x-values for the function, $y = W(4x)$.

The x-values for $W(x)$ will be ○ divided ○ multiplied by 4.

2. The functions $f(x)$ and $g(x)$ are shown on the graph, where $g(x) = -f(x)$.



Complete the sentence to describe the effect on the y-values for the function $y = f(x)$.

The y-values for $f(x)$ will be ○ divided ○ multiplied by -1.

13.7.5 Wrap-Up: Ticket Out the Door

1. Determine how the values in a table of values for $K(x)$ and $N(x)$ will change given that $G(x) = x^2$. Complete the table.

Function	Values that Changed from $G(x) = x^2$	Description of the Change
$K(x) = (3x)^2$	○ x-values ○ y-values	○ Divided by _____ ○ Multiplied by _____
$N(x) = \frac{2}{3}x^2$	○ x-values ○ y-values	○ Divided by _____ ○ Multiplied by _____